

DR. BABASAHEB AMBEDKAR TECHNOLOGICAL UNIVERSITY, LONERE**Regular/Supplementary Winter Examination – 2024****Course: B. Tech****Branch: Electronics and Computer Engineering/ Electronics and Computer Science****Semester :3rd****Subject Code & Name: BTESC305 & Digital Electronics and Microprocessor****Max Marks: 60****Date:21/02/2025****Duration: 3 Hr.****Instructions to the Students:**

1. Each question carries 12 marks.
2. Question No. 1 will be compulsory and include objective-type questions.
3. Candidates are required to attempt any four questions from Question No. 2 to Question No. 6.
4. The level of question/expected answer as per OBE or the Course Outcome (CO) on which the question is based is mentioned in () in front of the question.
5. Use of non-programmable scientific calculators is allowed.
6. Assume suitable data wherever necessary and mention it clearly.

		(Level/CO)	Marks
Q. 1	Objective type questions. (Compulsory Question)		12
1	Which of the following is a type of digital logic circuit? a. Combinational logic circuits b. Sequential logic circuits c. Both Combinational & Sequential logic circuits d. None of these		1
2	Any signed negative binary number is recognized by its a. MSB b. LSB c. Nibble d. Byte		1
3	Convert the hexadecimal number (1E2) ₁₆ to decimal. a. 482 b. 484 c. 481 d. 483		1
4	Which of the following is a universal gate a. AND b. OR c. NOT d. NAND		1
5	What is the group of 1s in 4 cells of a K – map called? a. Pair b. Quad c. Octoate d. None of these		1
6	Ripple counters are also called a. VLSI counters b. SSI counters c. Asynchronous counters d. Synchronous counters		1
7	What kind of operation occurs in a J – K flip flop when both inputs J and K are equal to 1? a. Toggle operation b. Preset operation c. Reset operation d. Clear operation		1
8	Which of the following is a non-vectored input? a. INTR b. TRAP c. RST 7.5 d. RST 6.5		1
9	How many flags are there in an 8085 microprocessor? a. 4 b. 5 c. 6 d. 7		1

10	Byte contains __ number of bits		1
	a. 4 b. 8 c. 16 d. 32		
11	Which of the following is not a valid register pair in 8085 microprocessors?		1
	a. BC b. DE c. HL d. AB		
12	8086 microprocessor is a __ bit processor		1
	a. 4 b. 8 c. 16 d. 32		
Q. 2	Solve the following.		12
A)	Draw the symbol & write truth tables of the following logic gates (i) NAND (ii) NOR (iii) EX-OR		6
B)	State & Prove De-Morgan's theorem		6
Q.3	Solve the following.		12
A)	Compare Multiplexer & Demultiplexer		6
B)	Minimize & implement using K-Map $F(A, B, C, D) = \sum m(1, 2, 6, 8, 10, 15) + d(0, 3, 5, 11)$		6
Q. 4	Solve Any Two of the following.		12
A)	Draw a neat diagram of SR Flip-flop & explain its operation		6
B)	Design a 3-bit UP counter(ripple) using JK flip-flop		6
C)	What is a shift register? What are its types & enlist its applications		6
Q.5	Solve Any Two of the following.		12
A)	Draw the timing diagram of OPCODE fetch machine cycle		6
B)	Perform the operation $7FH + 01H$ & indicate the status of the flags (8085 microprocessor)		6
C)	Explain all addressing modes of 8085 microprocessor with examples		6
Q. 6	Solve Any Two of the following.		12
A)	Draw a neat architecture of 8086 microprocessor & enlist its salient features		6
B)	Explain flag register of 8086 microprocessor		6
C)	Explain any two assembler directives		6

*** End ***